Regression Modeling

Building on the cleaned dataset and insights from P1, you will now build and evaluate regression models. You will apply linear and polynomial regression, engineer new features, tune your models, and critically interpret the results. If your P1 target is categorical, define a continuous proxy target or clearly justify how you adapt the task.

Required Sections

Your notebook must contain the following sections **in order**.

1. Recap & Motivation

- One-paragraph summary of your problem and dataset (from P1)
- State the regression target clearly (which column, what it represents)
- Briefly explain what new understanding you bring from P1 that shapes your modelling choices

2. Feature Engineering

Go beyond the raw features from P1. You must include at least **three** of the following:

Technique	Example
Polynomial features	`price²`, `area × rooms`
Log / power transform	`log(income)` to reduce skew
Binning	Age → age group dummy
Interaction term	`distance_to_center × floor_count`
Aggregation	Mean price per neighbourhood
Datetime decomposition	Hour of day, day of week
Encoding	One-hot or ordinal for categorical features

For each new feature: explain what it represents and why you expect it to help.

3. Train / Validation / Test Split

- Split the data into **train (60%), validation (20%), test (20%)** - or use **train (80%) + test (20%) with cross-validation**
- Explain your choice
- Apply feature scaling (StandardScaler or MinMaxScaler) **after** splitting to prevent data leakage
- Report class/target distribution in each split (histogram of the target)

4. Baseline Model

- Fit a **simple linear regression** (one most-predictive feature only)
- Report Train R^2, Validation R^2, RMSE, MAE
- Plot predicted vs. actual values
- This baseline sets the performance floor for the rest of your models

5. Multiple Linear Regression

- Fit a model using all selected features
- Report the same metrics as above
- Examine coefficients: which features have the largest positive/negative impact?
- Check residual plots (residuals vs. fitted, histogram of residuals, Q-Q plot)

6. Polynomial Regression

- Try at least **degree 2 and degree 3** polynomial expansions
- Plot train and validation error as a function of polynomial degree (bias-variance curve)
- Identify the degree with the best validation performance
- Comment on underfitting vs. overfitting

7. Regularization

- Apply **Ridge (L2)** and **Lasso (L1)** regression
- Use cross-validation to select the regularization strength α (e.g., `RidgeCV`, `LassoCV`)
- Compare regularized models against the plain multiple regression model
- For Lasso: report which features were zeroed out and why that makes sense (or not)

8. Model Comparison

Summarise all models in a single table:

Model	**Train R^2**	**Val R^2**	**Test R^2**	**RMSE (test)**	**MAE (test)**
Baseline (simple LR)					
Multiple LR					
Polynomial (best degree)					
Ridge					
Lasso					

- Select your **best model** and justify the choice
- Report final test-set performance for the chosen model only

9. Interpretation & Reflection

- What do the model results tell you about your problem?
- Are the predictions good enough to be useful? Why or why not?
- What are the main sources of error?
- How will these findings guide your choices in P3?

Submission Requirements

Item	Details
File	Single Jupyter Notebook (`.ipynb`)
Filename	`p2_regression_<student_id>.ipynb`
Code quality	Clean cells, no dead code, cells run top-to-bottom without errors
Visualizations	All plots must have titles, axis labels, and legends where applicable

GitHub Repository

- Use the same **public** GitHub repository
- Push your notebook to the `p2/` folder
- Update `README.md` to include a short summary of P2 findings.

Submit the **link to your GitHub repository** on the Jupyter Notebook you submit.